Maiqi Jiang

mjiang04@wm.edu (757) 875-4331 maiqiverse.github.io MaiqiVerse maiqi-jiang-103346273

Education

PhD **College of William & Mary**, Computer Science — Williamsburg, VA
- Advisor: Prof. Yanfu Zhang — Aug 2025 – present

M.Sc. (with Distinction Award) **The Hong Kong Polytechnic University (QS Top 60)**, Information Technology — Hong Kong, Aug 2021 – Jan 2023

B.S. **Northeastern University, China (985 Project)**, Biomedical Engineering — Shenyang, China, Sept 2017 – June 2021

Experience

William & Mary, PhD Researcher — Federated Tuning & Structural Pruning for LLMs (SFT-P) — Williamsburg, VA, Aug 2025 – Jan 2026, 6 months
- Advisor: Prof. Yanfu Zhang
- Proposed **SFT-P**: FedAvg-based training that jointly learns client-specific structured pruning masks and adaptation for on-device deployment under non-IID private data; supports mixed client budgets.
- Implemented a client-conditioned mask generator (shared hypernetwork + private client embedding) producing hard binary routing masks via Bernoulli gating, and a residual-safe route/merge interface for dense-kernel execution; achieved **+8.5 Avg over the best federated baseline on LLaMA-7B at 50% pruning**. Submitted to **NAACL 2027**.

William & Mary, PhD Researcher — In-Context Knowledge Editing for LLMs (DR-IKE / FWE-IKE / MO-IKE) — Williamsburg, VA, Aug 2025 – present, 1 year 3 months
- Advisors: Prof. Yanfu Zhang, Prof. Haipeng Chen
- Main contributor to a research line on **retrieval-based in-context knowledge editing** for black-box LLMs, spanning policy-optimized retrieval, confidence-aware fusion, and constrained multi-objective RL.
- **DR-IKE** (EMNLP 2025, co-first): trained a BERT retriever via REINFORCE with a learnable threshold to dynamically size demonstrations. [code]
- **FWE-IKE** (ACL 2026 Findings): confidence-gated ensemble of LLM/BERT/MLP rankers with logit-space fusion, closing ~53% of the gap to the oracle on COUNTERFACT.
- **MO-IKE** (EMNLP 2026 Findings): formulated prompt construction as a Constrained MDP with Lagrangian relaxation and GRPO, jointly optimizing reliability, generality, and specificity.

William & Mary, PhD Researcher — Meta-Path Attention Reliability in Heterogeneous GNNs (MetaXplain) — Williamsburg, VA, Aug 2025 – Feb 2026, 7 months
- Advisor: Prof. Yanfu Zhang
- Conducted a controlled empirical study of whether meta-path attention is a **faithful explanation**, identifying regimes of alignment vs. statistically significant decoupling across multiple datasets and HGNN backbones.
- Introduced **MetaXplain**, a meta-path-aware post-hoc analysis protocol (view-factorized explanations, schema-valid perturbations, fusion-aware attribution), and proposed **MP-AEA** to quantify attention–explanation alignment; released an explainability benchmark codebase ExplainerForHGNN for the community. Submitted to **TMLR** (under review).

William & Mary, PhD Researcher — Biomedical HIN Link Prediction (IM-HIN / PAIRS) — Williamsburg, VA

- Advisor: Prof. Yanfu Zhang — Aug 2025 – Feb 2026
- Proposed **PAIRS**: framed protein-phenotype association prediction as **link prediction** over a biomedical heterogeneous information network, rather than per-label classification. — 7 months
- Built **IM-HIN** by integrating STRING, HumanNet XN, GeneMania (protein-protein interactions) and HPO (protein-phenotype associations) into a single graph; two type-specific encoders (protein encoder + phenotype encoder) feed a relation-prediction head.
- Accepted at **ICASSP 2026** (first author).

William & Mary, Project Lead — In-Context Task Learning: Reading Out a Kernel Classifier — Williamsburg, VA

- Advisor: Prof. Yanfu Zhang — Feb 2026 – present
- Identified **mapping-sensitive attention heads** through paired intact/randomized-label contrasts and characterized their outputs as a **kernel classifier** combining query-dependent retrieval with contextual class votes. On **TREC-fine** with Llama-3.1-8B and 180 demonstrations, direct readout from **8 selected heads** achieves **50.3% accuracy**, versus 51.6% for the full model. — 9 months
- Proposed **Selective TL Memory**, reusing cached demonstration K/V through selected heads with live query-dependent retrieval. Selective access improves the same fixed readout across **all 18 natural-language settings** at both tested head budgets; the **30-head readout** outperforms all compared adapted-vector baselines in **14/18 settings**. Submitted to **ICLR 2027** (under review).

William & Mary, Project Mentor — DyV2X: Dynamic Interaction Graphs for V2X Cooperative Trajectory Prediction — Williamsburg, VA

- Advisor: Prof. Yanfu Zhang — Dec 2025 – Mar 2026
- Mentored two interns (the first two authors); guided the design of **DyV2X**, a cooperative trajectory prediction framework that constructs per-timestep dynamic interaction graphs and uses a temporal graph attention encoder to jointly model evolving inter-agent interactions and individual kinematic history. — 4 months
- On V2X-Seq-TFD (9,320 target agents), improved over the V2X-Graph baseline on all metrics: **ADE** 1.110→0.952 (**14.2%↓**), **FDE** 1.894→1.664 (**12.1%↓**), **MR@2m** 27.9%→24.7%.
- Largest gains on safety-critical turning maneuvers (e.g., *left turn*: FDE 4.216→2.646, **37.2%↓** MR@2m 64.2%→47.5%). Submitted to **ICRA 2027** (under review).

William & Mary, Project Mentor — Intern-led Projects on LLM Efficiency & Long-Context — Williamsburg, VA

- Advisor: Prof. Yanfu Zhang — Sept 2025 – present
- **Domain-Adaptive Pruning (CDAM).** Mentoring *Noman Ali* on a **training-free activation measure** combining domain-specific and general-purpose statistics with feature-wise cross-domain interactions, integrated into Wanda and CFSP. On Llama-2-13B at **50% sparsity** with 1,024 calibration examples, CDAM raises CFSP's average biomedical accuracy from **34.45% to 38.70%** versus naive calibration mixing, while improving general-task accuracy from 58.73% to 61.32%. Targeting **ACL 2027**. — 1 year 2 months
- **ICL-Preserving Pruning (ICLPrune).** Mentoring *Chongyuan Sun* on improving pruned-LLM **in-context learning** performance under task-shifted prompts via demonstration-aware calibration (*ICLPrune: Improving In-Context Learning in Structured LLM Pruning via Demonstration-Aware Calibration*, targeting **ACL 2027**).
- **Long-Context & ICL.** Proposed the research direction and co-supervised (with Prof. Zhang) *Yiran Ding* on the **cross-lingual gap in long-context reasoning** under context-extension methods, leading to *One Direction Away: The Cross-Lingual Gap in Long-Context Reasoning and a Shift That Closes It* (submitted to **NAACL 2027**); mentoring *Muhammed Muminul Hoque* on demonstration-ordering policies for ICL motivated by the **Lost-in-the-Middle** effect, given per-example rankings.

Independent / Open Source, Project Lead — lmdiff: Behavioral Diagnostics for LLM Configurations — Williamsburg, VA; Oct 2025 – present; 1 year 1 month

- Advisor: Prof. Yanfu Zhang
- Built and released **lmdiff** (PyPI `lmdiff-kit`, GitHub), a Python library for comparing LLM **configurations**—weights + system prompt + decoding + adapter—via behavioral distance and multi-level diagnostics, rather than aggregate benchmark scalars.
- Centered the API on a `family()` abstraction that compares one base configuration against N variants on a shared set of probe-domain axes, treating weight modifications (fine-tuning, RLHF, adapters), runtime modifications (system prompt, decoding, ICL context), and quantization/pruning as first-class variants. Geometry-level metrics (per-domain drift, share, raw + selective cross-variant cosine) quantify *where* each variant moves and whether different modifications push in the same direction. Showcase 7-variant Llama-2-7B family reveals a system prompt can reshape next-token distributions more than 7B→13B scaling.
- Engineering: engine layer with HF Transformers + mock backends, engine reuse that keeps peak VRAM at base + 1 active variant in 7-variant families, lm-eval-harness multi-task probe loader with task→domain mapping, JSON schema versioning with auto-recompute on load, 5-channel reporting (ANSI / PNG / HTML / Markdown / JSON), ~860 tests. Active development; representation, trajectory, and causal metrics planned for v0.4+.

The Hong Kong Polytechnic University, Researcher — Automated Heterogeneous Network Learning (AutoGNR) — Hong Kong; Nov 2021 – Feb 2023; 1 year 4 months

- Advisor: Prof. Xiao Huang
- Co-developed **AutoGNR**, using differentiable NAS to automatically select heterogeneous aggregation paths and node-type combinations—conceptually aligned with structured pruning and bi-level optimization. [AutoGNR] [HINBaselines]
- Designed a non-recursive message-passing scheme that decouples path search from depth, improving search efficiency on large heterogeneous graphs.

City University of Hong Kong, Research Assistant — Temporal User Engagement Modeling in Polarized Social Platforms — Hong Kong; Feb 2023 – Dec 2023; 11 months

- Advisor: Prof. Junming Liu
- Fused image/video content features with text and interaction context via a temporal GNN for forecasting user engagement across polarized communities on Twitter/Reddit.
- Built multimodal feature pipelines and longitudinal datasets for studying engagement dynamics under polarization. [RedditProcess] [RedditDataset]
- Submitted as a short paper to **ICIS 2023**, proposing a counter-party graph attention network with sentiment diffusion for online engagement prediction.

Neusoft Inc., Software Developer Intern — Shenyang, China; Sept 2020 – Sept 2020; 1 month

Northeastern University, China, Undergraduate Researcher / Bachelor's Thesis — Semi-supervised Learning for Whole-Slide Pathology Images (WSI) — Shenyang, China; Nov 2018 – June 2021; 2 years 8 months

- Advisor: Prof. Xiaoyu Cui
- Developed weakly-/semi-supervised pipelines for gigapixel WSI using eye-tracking signals, focusing on efficient patch sampling and large-scale image processing. [Modeling] [Processing]
- Maintained reusable WSI tooling (tiling, feature extraction, dataset management) used across the lab's projects.

Publications

PAIRS, Not Labels: Predicting Protein-Phenotype Associations via Link Prediction — Apr 2026
Maiqi Jiang, Yanshuo Chen, Guodong Liu, Avinash Sahu, Ye Gao, Yanfu Zhang

Dynamic Retriever for In-Context Knowledge Editing via Policy Optimization (DR-IKE) Nov 2025
Mahmud Wasif Nafee, *Maiqi Jiang*, Haipeng Chen, Yanfu Zhang
aclanthology.org/2025.emnlp-main.848

Towards Reliable, Generalizable, and Specific In-Context Knowledge Editing via Multi-Objective Reinforcement Learning (MO-IKE) Nov 2026
Xuzhong Wang, Maiqi Jiang, Tejal Nair, Girija Bhusal, Yanfu Zhang, Haipeng Chen

Confidence-Aware Ranker Ensembles for Robust In-Context Knowledge Editing (FWE-IKE) July 2026
Tejal Nair, Mahmud Wasif Nafee, Maiqi Jiang, Ashley Gao, Haipeng Chen, Yanfu Zhang

Hierarchical Convolution Multibranch Transformer for EEG Signals Apr 2026
Omid Mersa, Maiqi Jiang, Ye Gao, Qun Li, Yanfu Zhang

In-Context Task Learning: Reading Out a Kernel Classifier Sept 2026
Maiqi Jiang, Xugui Zhou, Bin Ren, Guodong Liu, Yanfu Zhang

SFT-P: Federated Tuning and Pruning for LLMs Oct 2026
Maiqi Jiang, Xugui Zhou, Bin Ren, Junyi Li, Shangqian Gao, Yanfu Zhang

One Direction Away: The Cross-Lingual Gap in Long-Context Reasoning and a Shift That Closes It Oct 2026
Yiran Ding, Maiqi Jiang, Qun Li, Yanfu Zhang

Representation Change and Adaptive Stopping in In-Context Learning Oct 2026
Tejal Nair, Omid Mersa, Zhen Xu, Maiqi Jiang, Xuzhong Wang, Ashley Gao, Gang Zhou, Haipeng Chen, Yanfu Zhang

Is Meta-Path Attention an Explanation? Evidence of Alignment and Decoupling in Heterogeneous GNNs Feb 2026
Maiqi Jiang, Noman Ali, Yiran Ding, Yanfu Zhang
openreview.net/forum?id=g9cnIFAYB3

Disentangling Memorization in Language Models via Sparse Representation Intervention Sept 2026
Jiarui Li, Maiqi Jiang, Junyi Li, Bin Wang, Shangqian Gao, Yanfu Zhang

DyV2X: Dynamic Interaction Graphs for V2X Cooperative Trajectory Prediction Sept 2026
Yiran Ding, Muhammed Muminul Hoque, Maiqi Jiang, Yongming Qin, Sidi Lu, Yanfu Zhang

Automated Heterogeneous Network Learning with Non-Recursive Message Passing (AutoGNR) Jan 2025
Zhaoqing Li, *Maiqi Jiang*, Shengyuan Chen, Bo Li, Guorong Chen, Xiao Huang
arxiv.org/abs/2501.07598

Teaching

- TA — Topological Data Analysis (CSCI 520/420), William & Mary, Spring 2026
- TA — Neural Networks for Machine Learning (CSCI 446/546), William & Mary, Fall 2025

Technical Skills

LLM Training & Adaptation: Federated LLM tuning with FedAvg; structured pruning with learned binary masks and dense-kernel route/merge execution (LLaMA-7B); HuggingFace + PyTorch.

RL for LLMs: REINFORCE-based and GRPO-based retrieval / policy optimization for in-context knowledge editing (DR-IKE, MO-IKE); reward design under Constrained MDP with Lagrangian relaxation.

Graph Learning: Heterogeneous GNNs (HAN/HGT-family), temporal graph networks (TGN / DyGFormer), differentiable NAS over heterogeneous aggregation paths; PyG-style implementation.

Mechanistic Interpretability: Attention head analysis for ICL; sparse autoencoder (SAE) representation interventions with ReFT; meta-path attention faithfulness diagnostics.

Systems & Tooling: Python library design and PyPI release (`lmdiff-kit`); multi-codebase open-source benchmarks (ExplainerForHGNN, HINBaselines); large-scale data pipelines for WSI and multimodal social media; mentoring code reviews across 4 intern projects.

Languages: Python, PyTorch, C++. Spoken: Chinese (native), English (fluent).

Research Interests

- What interventions actually change inside an LLM — structured pruning, knowledge editing, adapters, decoding
- Building interventions that preserve the capabilities that matter (in-context learning, domain knowledge, factual reliability)
- Mechanistic analysis of in-context learning and of attention-based explanations
- RL-based methods for LLMs (policy-optimized retrieval, constrained multi-objective RL)
- Federated / on-device LLM adaptation and model compression
- Behavioral diagnostics and released tooling for LLM configurations
- Graph learning and explainability (heterogeneous GNNs, biomedical HINs, V2X)

Service

- Research Assistant: City University of Hong Kong, with Prof. Junming Liu (Feb 2023 – Dec 2023)
- Software Developer Intern: Neusoft Inc., China (Sep 2020)
- PC member / Reviewer: **ICASSP 2026**, **NeurIPS 2026**, **CIKM 2026 Workshop**, **ICLR 2027**, **ACM Transactions on Computing for Healthcare**
- Author & maintainer of lmdiff-kit (PyPI)
- Open Source Contributor: The Unified AI Framework (Jul 2023); ExplainerForHGNN benchmark (2025–2026)